Square density → Sinc

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CHAPTER #4

4.2.2 POWER DENSITY SPECTRUM OF PERIODIC SIGNALS

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$ (avg. of power of discrete time)

P_x in terms of fourier:

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} x(n)\, x^*(n)$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x(n) \left(\sum_{k=0}^{N-1} c_k^* e^{-j2\pi kn/N} \right)$$ c_k - individual freq. component.

$$= \sum_{k=0}^{N-1} c_k^* \underbrace{\left[\frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \right]}_{c_k}$$

$$= \sum_{k=0}^{N-1} |c_k|^2$$

* $$P_x = \frac{1}{N} \sum_{k=0}^{N-1} |x(n)|^2$$ (Parseval's theorem)

Energy of the signal $x(n)$:

$$E_N = \sum_{n=0}^{N-1} |x(n)|^2$$ in terms of $x(n)$

$$E_N = N \sum_{k=0}^{N-1} |c_k|^2$$ in terms of c_k.

EXAMPLE #4.2.2

4.2.3 THE FOURIER TRANSFORM OF DISCRETE TIME APERIODIC SIGNAL:

APERIODIC → repeated signal

$$X(w) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-jwn}$$ (fourier transform of finite energy discrete time signal)

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$X(\omega)$ is periodic with 2π

$$X(\omega + 2\pi k) = \sum_{n=-\infty}^{\infty} x(n) e^{-j(\omega+2\pi k)n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} e^{-j2\pi k n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$= X(\omega)$ - exist after every $2\pi k$

* SYNTHESIS EQUATION , FOURIER TRANSFORM:

$$x(n) = \frac{1}{2\pi}\int_{2\pi} X(\omega) e^{j\omega n}\, d\omega \quad , \quad X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

4.2.5 ENERGY DENSITY SPECTRUM OF APERIODIC

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\omega)|^2\, d\omega \quad \text{(Parseval's theorem)}$$

in terms of fourier transform

$S_{xx}(\omega) = |X(\omega)|^2$ (distribution of energy as a function of frequency).

20/2 Q1) Find the discrete time fourier transform of the following:

$$x(n) = \begin{cases} 1 & 0 \le n \le 4 \\ 0 & ; \text{ otherwise} \end{cases}$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

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$$X(\omega) = \sum_{n=0}^{4} \left(1 e^{-j\omega n} \right)$$

$$= 1 + e^{-j\omega} + e^{-j\omega 2} + e^{-j\omega 3} + e^{-j\omega 4}$$

$$X(\omega) = |e^{j\theta}|$$

Let's find, $|X(\omega)|$ & $e^{j\theta}$

$$x(n) e^{-j\omega n} = \sum_{n=0}^{4} e^{-j\omega n} = \sum_{n=0}^{N-1} (e^{-j\omega})^n \quad \because N=5$$

$$= \underbrace{\frac{1-e^{-j\omega 5}}{1-e^{-j\omega}}}_{\text{geometric summation}} = \frac{e^{-j\frac{\omega 5}{2}}}{e^{-j\omega/2}} \times \frac{e^{+j\frac{\omega 5}{2}} - e^{-j\frac{\omega 5}{2}}}{e^{\frac{j\omega}{2}} - e^{-\frac{j\omega}{2}}}$$

$$= e^{-2j\omega} \cdot \frac{\sin(5\omega/2)}{\sin(\omega/2)} = e^{j\theta} (|X(\omega)|$$

As,

$$e^{-j2w} = e^{j\theta}$$

Hence, $\angle X(w) = -2w$

$$|X(\omega)| = \frac{\sin(5\omega/2)}{\sin(\omega/2)}$$

Q2) Find the discrete time fourier transform of the following signal: $x(n) = \{ a^n u(n) \quad , |a| < 1$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} = \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

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$$= \frac{(ae^{-j\omega})^0}{1 - ae^{-j\omega}} = \frac{1}{1 - ae^{-j\omega}}$$

$\therefore \sum_{k=a}^{\infty} x^k = \frac{x^a}{1-x}$

$$= \frac{1}{1 - a(\cos\omega - j\sin\omega)}$$

$$X(\omega) = \frac{1}{1 - a\cos\omega + ja\sin\omega}$$

$$|X(\omega)| = \frac{1}{\sqrt{(\underbrace{1 - a\cos\omega}_{real})^2 + (\underbrace{a\sin\omega}_{imaginary})^2}}$$

$$= \frac{1}{\sqrt{1^2 + a^2\cos^2\omega - 2a\cos\omega + a^2\sin^2\omega}}$$

$$= \frac{1}{\sqrt{1 - 2a\cos\omega + a^2(\underbrace{\cos^2\omega + \sin^2\omega}_{=1})}}$$

$$= \frac{1}{\sqrt{1 - 2a\cos\omega + a^2}}$$

lets find minima & maxima (for plotting) by derivative

let, $g(\omega) = 1 - 2a\cos\omega + a^2$

$$\frac{dg}{d\omega} = 0 - 2a(-\sin\omega) + 0 = 0$$

$\therefore g(\omega) = 2a\sin(\omega) = 0$

$\sin(\omega) = 0$ $\quad$ $\because$ Sin is not 0 at $0, 2\pi, \ldots$

$$|X(\omega)| = \frac{1}{\sqrt{1 - 2a + a^2}} = \frac{1}{\sqrt{(1-a)^2}}$$

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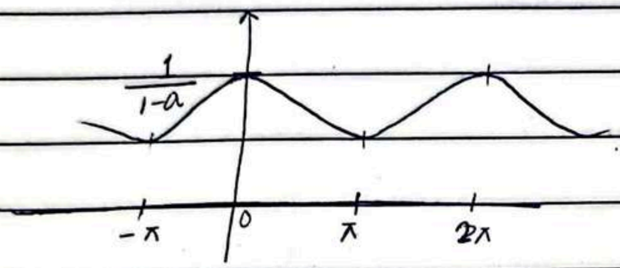
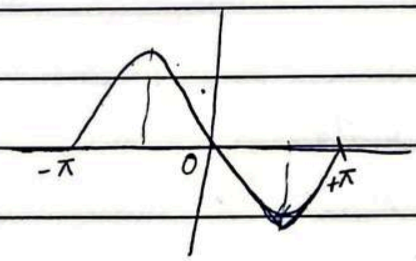
$$|X(0)| = \frac{1}{1-a}$$

$$|X(2\pi)| = \frac{1}{1-a}$$

$$|X(\pi)| = \frac{1}{\sqrt{1+2a+a^2}} = \frac{1}{\sqrt{(1+a)^2}} = \frac{1}{1+a}$$



$$\angle X(\omega) = \angle num - \angle den$$
$$= \angle 1 - \angle(1 - a\cos\omega + ja\sin\omega)$$
$$= \tan^{-1}(0/1) - \tan^{-1}(a\sin\omega / (1 - a\cos\omega))$$



EXAMPLE # 4.2.4

Determine the fourier transform & the energy density spectrum of the seq.

$$x(n) = \begin{cases} A, & 0 \le n \le L-1 \\ 0, & \text{otherwise} \end{cases}$$

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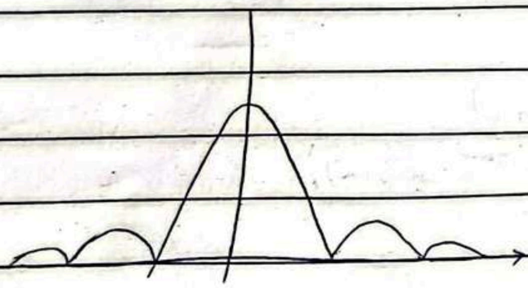
$$X(\omega) = \sum_{n=0}^{L-1} (A e^{-j\omega n})$$

$$= A + Ae^{-j\omega} + Ae^{-j\omega 2} \ldots Ae^{-j\omega(L-1)}$$

$$= A\left(\frac{1-e^{-j\omega L}}{1-e^{-j\omega}}\right)$$

$$X(\omega) = Ae^{-j\omega L/2(L-1)} \frac{\sin(\omega L/2)}{\sin(\omega/2)}$$

$$|X(\omega)| = \begin{cases} |A| L & ; \\ |A| \dfrac{\sin(\omega L/2)}{\sin(\omega/2)} & ; \end{cases}$$



* EXAMPLE #4.4.4 (a)

4/3 Find DTFT of the following signal:

$$x(n) = \begin{cases} 1 & ; \; 0 \leq n \leq 4 \\ 0 & ; \; \text{otherwise} \end{cases}$$

$$\therefore X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{4} 1 e^{-j\omega n}$$

$$= 1 + e^{-j\omega} + e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega}$$

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$X(\omega) = |e^{j0}|$

$x(n)e^{-j\omega n} = \sum_{n=0}^{4} e^{-j\omega n} = \sum^{N-1}$

EXAMPLE #4.4.4

$$y(n) = a y(n-1) + b x(n) \quad ; \quad 0 < a < 1$$

a) $h(n) = b a^n u(n)$ $\quad |H(\omega)| = ? \ , \ \angle H(\omega) = ?$

$$H(\omega) = \sum_{n=-\infty}^{\infty} h(n) e^{-j\omega n}$$

$$= \frac{b}{1 - a e^{-j\omega}}$$

$$|H(\omega)| = \frac{b}{1 - a(\cos\omega + j\sin\omega)}$$

$$|H(\omega)| = \frac{b}{\sqrt{(1 - a\cos\omega)^2 + (\sin\omega)^2}}$$

$$= \frac{b}{\sqrt{1 + a^2\cos^2\omega - 2a\cos\omega + a^2\sin^2\omega}}$$

$$= \frac{b}{\sqrt{1 - 2a\cos\omega + a^2(1)}}$$

$$\angle H(\omega) = \tan^{-1}\left(\frac{a\sin\omega}{1 - a\cos\omega}\right)$$

6/3 MULTIRATE DIGITAL SIGNAL PROCESSING

In many practical applications of digital signal processing one is faced with a problem of changing the sampling rate of a signal, either increasing it or decreasing it.

EXAMPLE #

- In telecommunication systems that transmit and receive different type of signals (e.g. speech, videos, etc) there is a requirement to process the various signals at different rates according to the different bandwidths of the signals at the process of converting the signal from a given rate to a different rate is called 'sampling rate conversion'.
- AUDIO PROCESSING:
  Audio signals can be converted from 44.1kHz. (for CD's) to 48kHz (for DVD's)
- Image Processing:
  Images/videos can be upscaled to higher resolution.
- SEISMIC DATA PROCESSING:
  The size of seismic data, can be reduced for storage and transmission.
- REDUCED BANDWIDTH:
  By reducing the sampling rate of the signal (decimation) the amount of data that needs to be transmitted/stored is decreased. This directly reduces the required bandwidth.

$$\xrightarrow[F_x = \frac{1}{T_x}]{x(n)} \boxed{\text{SYSTEM}} \xrightarrow[F_y = \frac{1}{T_y}]{y(n)}$$

$$F_x \neq F_y$$

Sampling rate conversion of a digital signal can be accomplished in one of two general methods. One method

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is to pass the digital signal through a D/A converter, filter it if necessary, and then to resample the resulting analog signal at the desired rate. The second method is to perform the sampling rate conversion entirely in the digital domain.

i) $\xrightarrow[F_x = \frac{1}{T_x}]{x(n)}$ D/A CONVERTER $\longrightarrow$ SAMPLER $\xrightarrow[F_y = \frac{1}{T_y}]{Y(n)}$ (first method)

$T_x \neq T_y$

PROBS & CONS:

- The benefit of this method is that F_x & F_y can have any arbitrary value.
- The problem in this method is extensive approximation, which distorts the signal.

ii) DIGITAL DOMAIN METHOD:

- In this method, sampling rate conversion is performed entirely in the digital domain.
- In this method, F_x & F_y will have same relationships. However, we need not to perform D/A conversion.

$$F_y = IF_x$$

UPSAMPLER:

$\xrightarrow[F_x = \frac{1}{T_x}]{x(n)}$ DISCRETE SYSTEM $\xrightarrow[F_y = \frac{1}{T_y}]{y(n)}$

$F_y > F_x$

Let,

$$x(n) = \{1, 2, 3, 4\}$$
$$n = 0, 1, 2, 3$$

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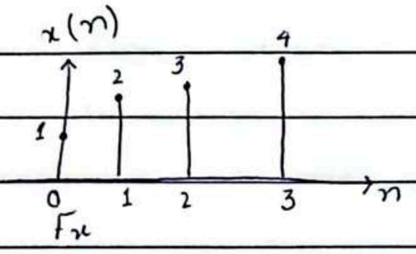
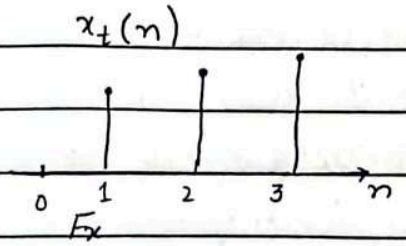


⇒



If sampling rate is increased, it means T_x is reduced and F_x is increased.

lets, assume we upsample the signal by a factor of 2.

$x(n) \rightarrow \boxed{\uparrow 2} \rightarrow y(n)$ ⇒ $\xrightarrow{F_x} \boxed{\uparrow I} \rightarrow F_y$

$F_y = 2 F_x$ $\qquad$ $F_y = I F_x$

$(I-1)$ zeros will be added between points.

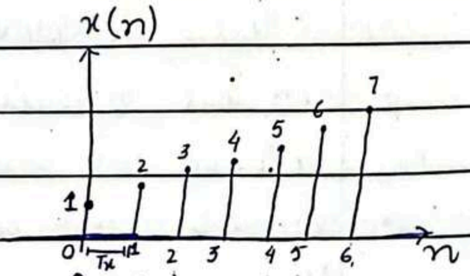
11/3 DOWN SAMPLER: (effecient storage)

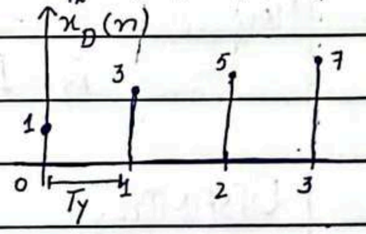
$F_x \rightarrow \boxed{\downarrow 2} \rightarrow F_y$

$F_x > F_y$

let,

$x(n) = \{1, 2, 3, 4, 5, 6, 7\}$
$n = 0, 1, 2, 3, 4, 5, 6$





Decimation by factor D.

$\xrightarrow{F_x} \boxed{\downarrow D} \xrightarrow{F_y}$

$F_y = F_x / D$

$(D-1)$ samples will be dropped.

no. of samples ↓
≡ processing ↑

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CHAPTER # 5

DISCRETE FOURIER TRANSFORM (DFT)

(if $X(\omega)$ is discrete - DFT)
(if $X(\omega)$ is continuous - DTFT)

Frequency analysis of a discrete time signals is usually performed on a digital signal processing. To perform frequency analysis on a time signal $\{x(n)\}$, we convert the time domain sequence to an equivalent frequency domain representation. We know that such a representation is given by the fourier transform $X(\omega)$ of the sequence $x(n)$. However, $X(\omega)$ is continuous function of frequency and therefore it is not a computationally convienient representation of the sequence $\{x(n)\}$, and therefore, we now consider the representation of a sequence $x(n)$ by samples of its spectrum $X(\omega)$. Such representation is called discrete fourier transform (DFT).

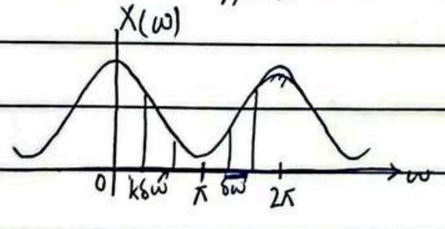
5.1 FREQUENCY-DOMAIN SAMPLING: The discrete fourier transform

5.1.1 FREQUENCY DOMAIN SAMPLING OF DISCRETE-TIME SIGNALS:

Recall that

sampling frequency ← I/p no. of samples

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \qquad (5.1.1)$$



δω - distance b/w two samples.

Suppose that we sample $X(\omega)$ periodically in frequency at spacing of $\delta\omega$ radians between successive samples, where $X(\omega)$ is periodic with period 2π.

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⇒ For convenience, we take N equidistant samples in the interval $0 \le \omega < 2\pi$ (after 2π, it repeats) with spacing $\delta\omega = 2\pi/N$, as shown in figure.

⇒ If we evaluate (5.1.1) at $\underbrace{\omega = 2\pi k/N}_{\text{equidistant samples}}$ we obtain,

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\frac{2\pi k}{N}} \quad ; k = 0, 1, \ldots N-1$$

13/3 DFT: $X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi k}{N}}$ (i) ; $k = 0, 1, 2, \ldots N-1$

IDFT: $x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi k}{N}}$ (ii) ; $n = 0, 1, 2, \ldots N-1$

THE DFT AS A LINEAR TRANSFORMATION:

The Formulas in eq (i) and eq (ii) may be expressed as

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad ; k = 0, 1, 2, \ldots N-1$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(k) W_N^{-kn} \quad ; k = 0, 1, 2, \ldots N-1$$

where, $W_N = e^{\frac{j2\pi}{N}}$

EXAMPLE # 5.1.3

Calculate

Q. 4-POINT DFT:

$N = 4$, $k = n = 0:3$

⇒ For

$k = 0$

$$\Rightarrow \underbrace{x(0) W_4^0 + x(1) W_4^0 + x(2) W_4^0 + x(3) W_4^0}_{x_N \Rightarrow 4\times1}$$

⇒ $k = 1$

$$\Rightarrow x(0) W_4^0 + x(1) W_4^1 + x(2) W_4^2 + x(3) W_4^3$$

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$k = 2$

$$\Rightarrow x(0)W_4^0 + x(1)\overset{kn=2(1)}{W_4^2} + x(2)\overset{kn=2(2)}{W_4^4} + x(3)W_4^6$$

$k = 3$

$$\Rightarrow x(0)W_4^0 + x(1)W_4^3 + x(2)W_4^6 + x(3)W_4^9$$

$$\underset{4\times4}{W_N}\,\underset{4\times1}{x_N} = X(k)_{4\times1}$$

4 point DFT

$$W_N = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}_{4\times4} \quad \text{(iii)}$$

(∵ $4\times4 \Rightarrow 4$ - Point)

(if 5-Point $\Rightarrow 5\times5$)

$W_4^0 = ?$

$W_4^1 = ?$

$W_4^2 = ?$

$W_4^3 = ?$

$$W_4^0 = W_4^{kn} = e^{\frac{-j2\pi(0)}{4}} = 1$$

$$W_4^1 = W_4^{kn} = e^{\frac{-j2\pi(1)}{4}} = e^{-j\frac{\pi}{2}} = \cos(\pi/2) - j\sin(\pi/2)$$

$$= 0 - j(1) = -j$$

$$W_4^2 = W_4^{kn} = e^{\frac{-j2\pi(2)}{4}} = e^{-j\pi} = \cos(\pi) - j\sin(\pi)$$

$$= -1 - 0 = -1$$

$$W_4^3 = e^{\frac{-j2\pi(3)}{4}} = e^{-j\frac{3\pi}{2}} = \cos(3\pi/2) - j\sin(3\pi/2)$$

$$= 0 - j(-1) = j$$

By periodicity property

$W_4^0 = W_4^4$

$W_4^1 = W_4^5 = W_4^9$ ∵ $4+1=5$, ∵ $5+4=9$

$W_4^2 = W_4^6$

∵ Periodicity:

$$W_N^R = W_N^{R+N}$$

if Compute the DFT of the 4-point sequence:
$x(n) = [0 \quad 1] \Rightarrow 1\times 2 \Rightarrow [0 \quad 1 \quad 0 \quad 0]^T = 4\times 4$
& $W_N \Rightarrow 4\times 4$

* Periodic → Sequence repeat
Aperiodic → 2π do not repeat over any interval.

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⇒ (iii)
$$W_N = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

EXAMPLE # 5.1.3

Compute the DFT of the 4-point sequence
$x(n) = [0 \quad 1 \quad 2 \quad 3]$

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}_{4\times 4} \times \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}_{4\times 1} = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix} = X(k)$$

~~CIRCULAR SYMME~~

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20/3 CIRCULAR SYMMETRIES OF A SEQUENCE:

As, we have seen the N-point DFT of a sequence, $x(n)$ of length $N \geq L$ is equivalent to the N-point DFT of a periodic sequence $x_p(n)$, of period N, which is obtained by periodically extending $x(n)$ that is:

$$\underbrace{x_p(n)}_{\text{periodic copies of } x(n)} = \sum_{\ell=-\infty}^{\infty} x(n-\ell N) \quad \text{(repetition)}$$

Now suppose we shift the periodic sequence $x_p(n)$ by 'k' units to the right. Thus we obtained another periodic sequence

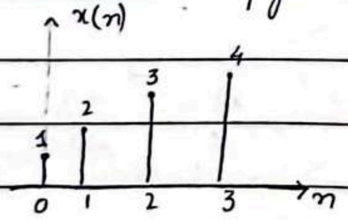
$$x'_p(n) = x_p(n-k)$$
$$= \sum_{\ell=-\infty}^{\infty} x(n-k-N\ell)$$

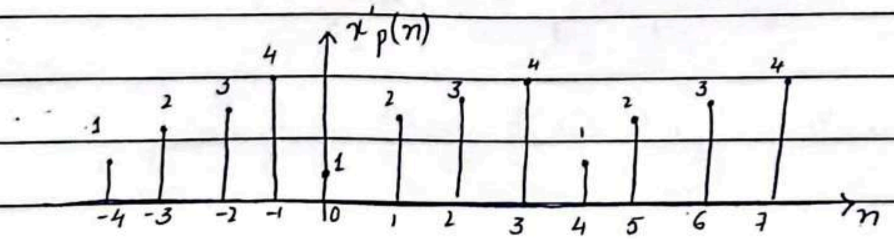
The finite duration sequence

$$x'(n) = \begin{cases} x'_p(n) & ; \quad 0 \leq n \leq N-1 \\ 0 & ; \quad \text{otherwise} \end{cases}$$

is related to $x(n)$ by a circular sequence.

As, shown in figure #1 for N=4





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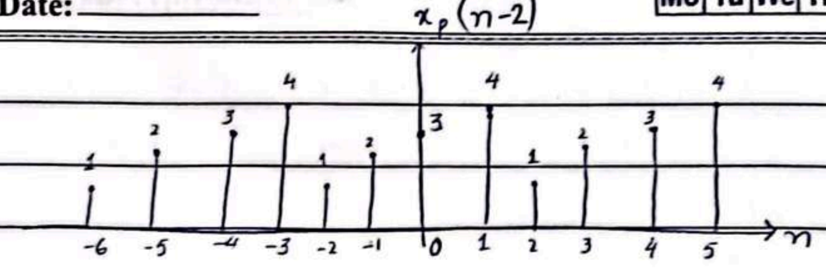

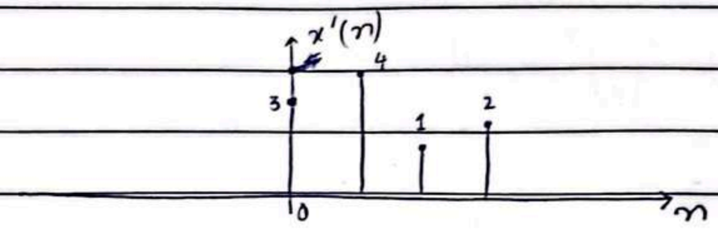

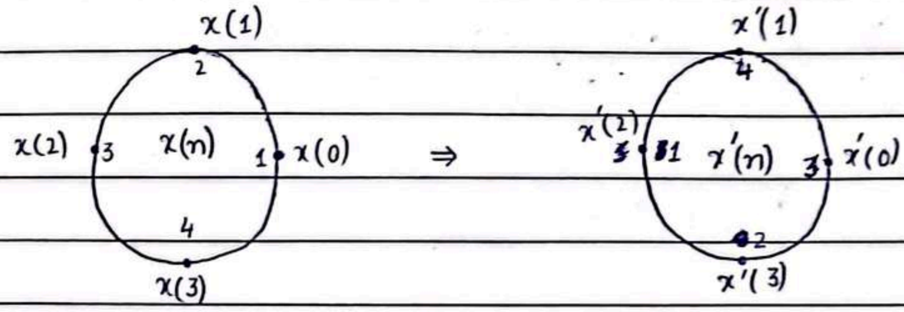

Circular Shift.

transform → spectral domain

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CHAPTER # 3
THE Z-TRASFORM

3.1

3.1.1 THE DIRECT Z-TRANSFORM

$$\underbrace{X(z)}_{\text{complex plane}} \equiv \sum_{n=-\infty}^{\infty} \underbrace{x(n)}_{\text{time domain signal}} z^{-n}$$

∵ z — complex variable

Z-transform of a signal x(n)

$$X(z) \equiv Z[x(n)]$$

* value of z is important, where series converges (finite duration)

EXAMPLE # 3.1.1

Determine z-transform

a) $x_1(n) = \{1, 2, 5, 7, 0, 1\}$

$$X_1(z) = x(1)z^{+0} + x(2)z^{-1} + x(3)z^{-2} + x(4)z^{-3} + x(5)z^{-4} + x(6)z^{-5}$$

$$= 1z^{-0} + 2z^{-1} + 5z^{-2} + 7z^{-3} + 0z^{-4} + 1z^{-5}$$

$$= 1 + \frac{2}{z} + \frac{5}{z^2} + \frac{7}{z^3} + 0 + \frac{1}{z^5}$$

ROC entire z-plane except z = 0

b) $x_2(n) = \{\overset{-2}{1}, \overset{-1}{2}, \underset{\uparrow}{\overset{0}{5}}, \overset{1}{7}, \overset{2}{0}, \overset{3}{1}\}$

$$X_2(z) = 1 \cdot z^{2} + 2z^{1} + 5z^{0} + 7z^{-1} + 0 + 1z^{-3}$$

ROC; entire z-plane except z = 0 & z = ∞

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CHAPTER #3

EXAMPLE # 3 1.2

Determine the z- transform:

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$x(n) = \left[1, \left(\frac{1}{2}\right), \left(\frac{1}{2}\right)^2, \left(\frac{1}{2}\right)^3 \cdots \left(\frac{1}{2}\right)^n \cdots\right]$$

$$X(z) = 1 + \frac{1}{2} z^{-1} + \left(\frac{1}{2}\right)^2 z^{-2} \cdots + \left(\frac{1}{2}\right)^n z^{-n} + \cdots$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n$$

infinite geometeric series:

$$1 + A + A^2 + A^3 + \cdots = \frac{1}{1-A} \quad ; \text{ if } |A| < 1$$

~~X(z) converges to:~~

$$\Rightarrow X(z) = \frac{1}{1 - (1/2) z^{-1}}$$

Region of convergence

ROC ; $|z| > 1/2$

EXAMPLE # 3.1.3

Determine the z- transform:

$$x(n) = a^n u(n) = \begin{cases} a^n & ; n \geq 0 \\ 0 & ; n < 0 \end{cases}$$

$$x(n) = \{0, 0, \underset{\uparrow}{a^0}, a^1, a^2, a^3 \cdots a^n\}$$

$\sum_{n=0}^{\infty} \rightarrow$ open form expression . ; because of ∞.

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$\cancel{X(z) = 1 + a + a^2 + a^3 + \dots} \; a^n$

$$X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

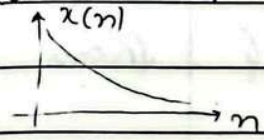
$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (a z^{-1})^n$$

If, $|az^{-1}| < 1$ or equivalent , $|z| > |a|$

$\Rightarrow$ Series converges to : $\dfrac{1}{1-az^{-1}}$; ROC : $|z| > |a|$

$$x(n) = a^n u(n) \xleftrightarrow{z} X(z) = \frac{1}{1-az^{-1}} \quad ; \text{ROC } |z| > |a|$$



EXAMPLE #3.1.4

$$x(n) = -a^n u(-n-1) = \begin{cases} 0 & ; n \geq 0 \\ -a^n & ; n \leq -1 \end{cases}$$

$$X(z) = \sum_{n=-\infty}^{-1} (-a^n) z^{-n}$$

$$= -\sum_{l=1}^{\infty} (a^{-l} z)^l \quad ; l = -n$$

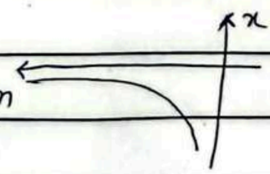
$= -(-\infty) = \infty$, $-(-1) = 1$

$\Rightarrow \sum_{1}^{\infty}$ closed form expression

using geometric series: $A + A^2 + A^3 + \dots = A(1 + A + A^2 \dots) = \underbrace{A/(1-A)}$; $|A| < 1$

$$X(z) = -\frac{a^{-1} z}{1 - a^{-1} z} = \frac{1}{1 - az^{-1}} \quad \text{ROC}: |z| < |a|$$

$|az^{-1}| < 1$



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RELATION B/W FOURIER & Z-TRANSFORM

$Z[x(n)] = X(z)$ → Complex plane

$$\sum_{n=-\infty}^{\infty} (x(n)) \cdot z^{-n} = X(z)$$

$$z = re^{j\omega} = \overbrace{r\cos\omega}^{real} + j\overbrace{\sin\omega}^{Imaginary}$$

$$\sum_{n} x(n)\,(re^{j\omega})$$

$$= \sum_{n} x(n)(e^{j\omega})^{n} \qquad \because \text{for } r=1$$

* EXAMPLE # 3.1.5

Q. Show that, $Z\{1\} = \dfrac{z}{z-1}$; $|z| > 1$

find $X(z) = ?$; $x(n) = 1$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$$= \sum_{n=0}^{\infty} (1)\, z^{-n}$$

$$= z^{-0} + z^{-1} + z^{-2} + \cdots z^{-n}$$

* $= (1 - z^{-1})^{-1}$ (using geometric sum)

$$= (1 - 1/z)^{-1}$$

$$= \left(\frac{z-1}{z}\right)^{-1} = \frac{z}{z-1}$$